

that results in upthrust

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Clockwise moments = Anticlockwise moments about the hinge

$$(10000x)(2.0) = (50 \times 9.81)(2.5) \text{ --- Eqn (1)}$$

$$(10000x')(2.0) = (40 \times 9.81)(5.0) + (50 \times 9.81)(2.5) \text{ --- Eqn (2)}$$

$$\text{Eqn (2) - Eqn (1), } x' - x = 0.0981 \text{ m}$$

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